

Experiment-01

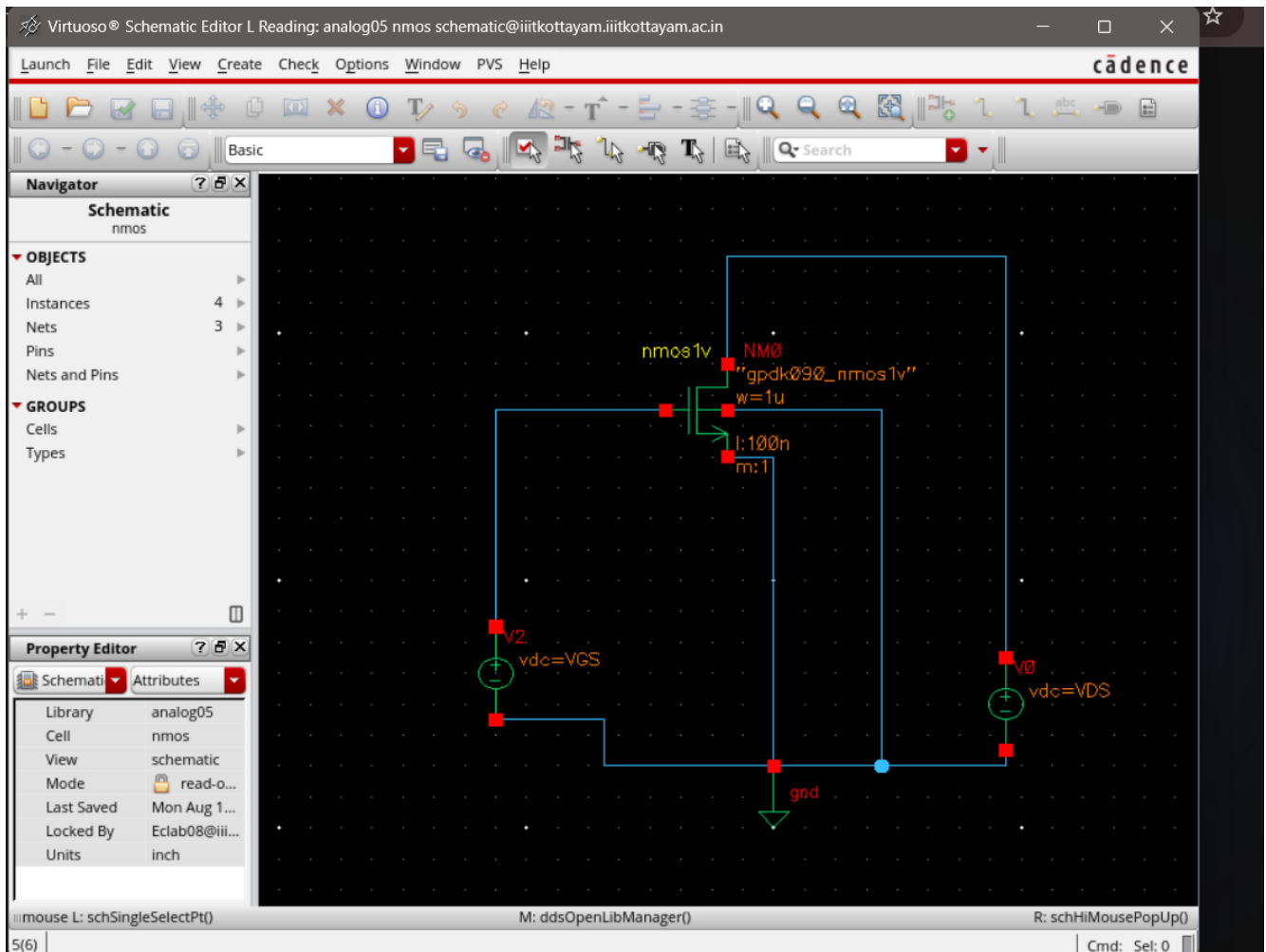
NMOS Characterization

Analog Electronics Lab

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GitHub	github.com/Subramanian-26/AnalogElectronicLab_2025BEC0005

Cadence Virtuoso / Spectre simulation

Schematic



NMOS test circuit used for characterization. The gate and drain are driven by the design variables VGS and VDS respectively, with the source connected to ground. The device shown has $W = 1 \mu\text{m}$ and $L = 100 \text{ nm}$.

element: writing instance parameter values to

Parametric Analysis - spectre(0): analog05 nmos schematic @iitkottayam.iitkottayam.ac.in

File Analysis Help

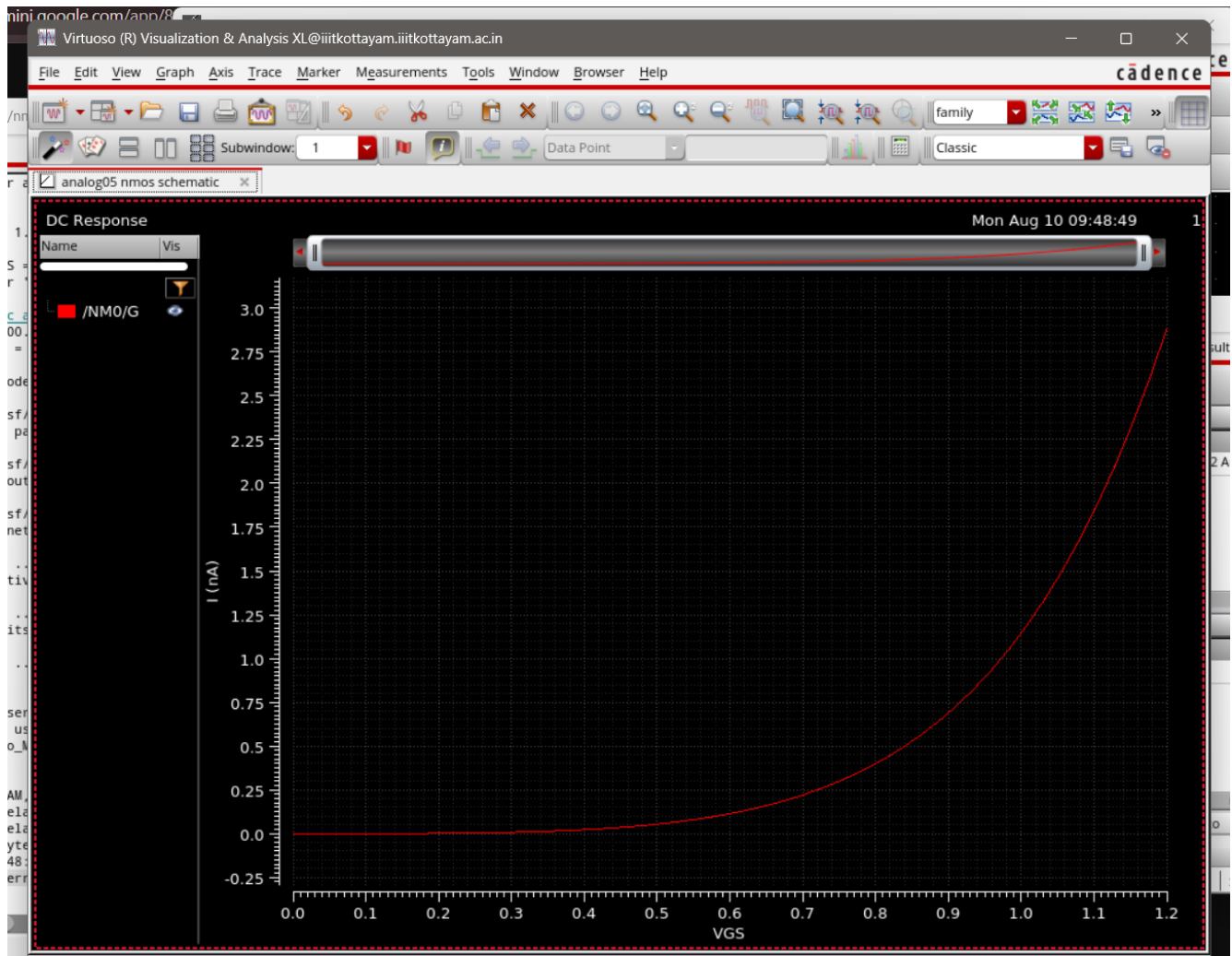
Ready

Run Mode: Sweeps & Ranges

Variable	Value	Sweep?	Range Type	From	To	Step Mode	Total Steps	Inclusion List	Exclusion List
VGS	1	☒	From/To	0	1.2	Auto			

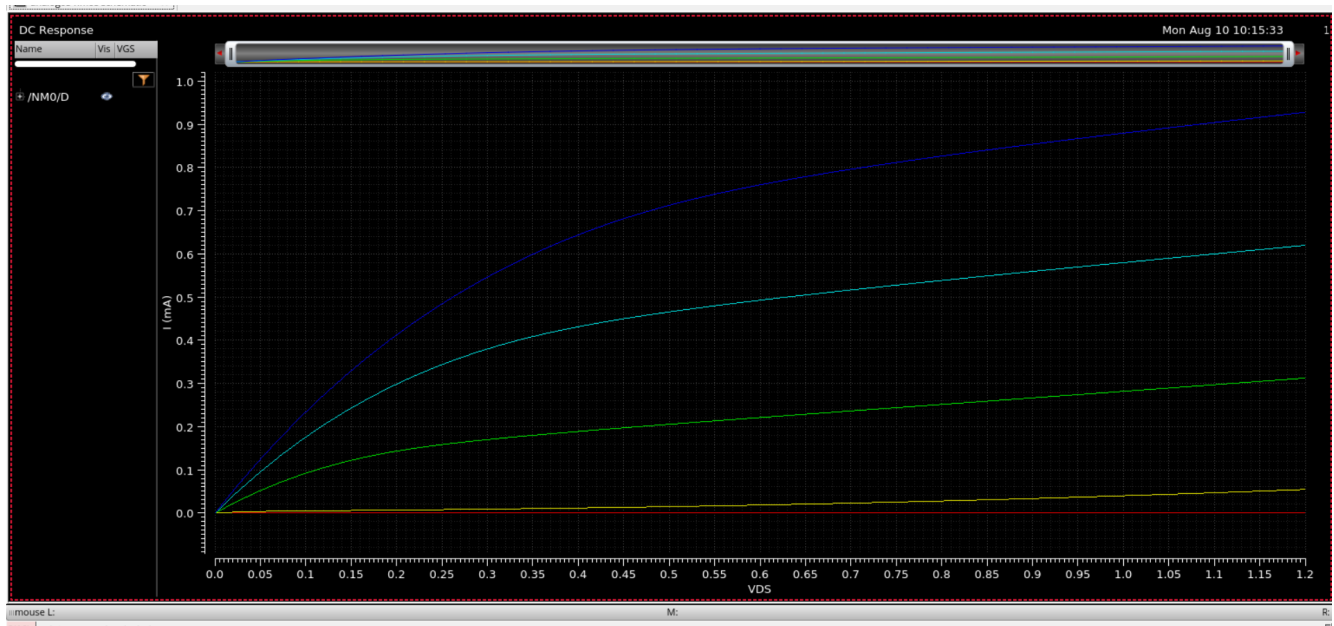
Parametric sweep setup for VGS: 0 V to 1.2 V.

VGS Transfer Characteristic



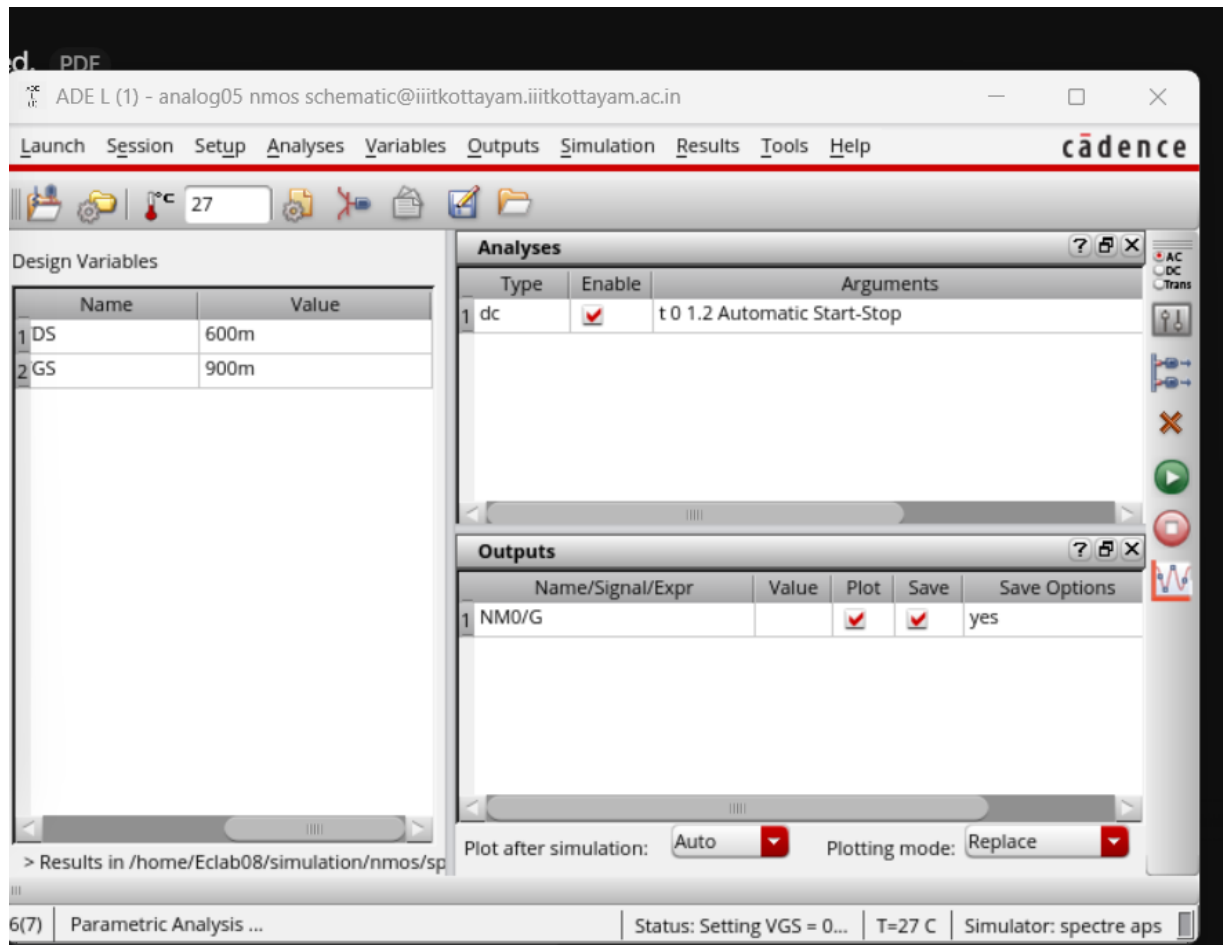
The DC response shows drain current increasing with V_{GS} . The current remains very small at low gate-source voltage and rises strongly as V_{GS} increases.

VDS Output Characteristics with Parametric Sweep

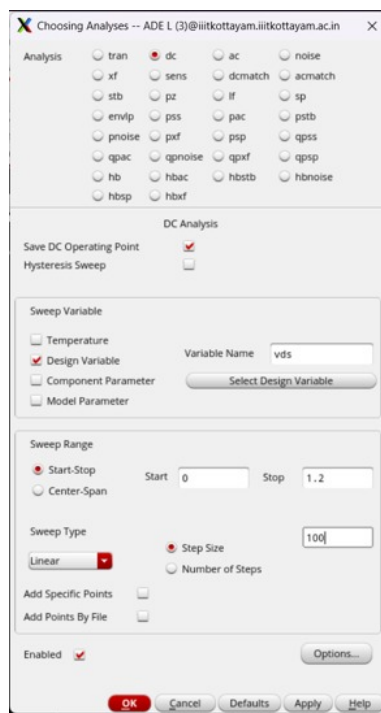


The output characteristics show I_D as a function of V_{DS} for multiple V_{GS} values. Increasing V_{GS} produces higher drain current, and the curves become less steep at larger V_{DS} values.

Simulation Setup

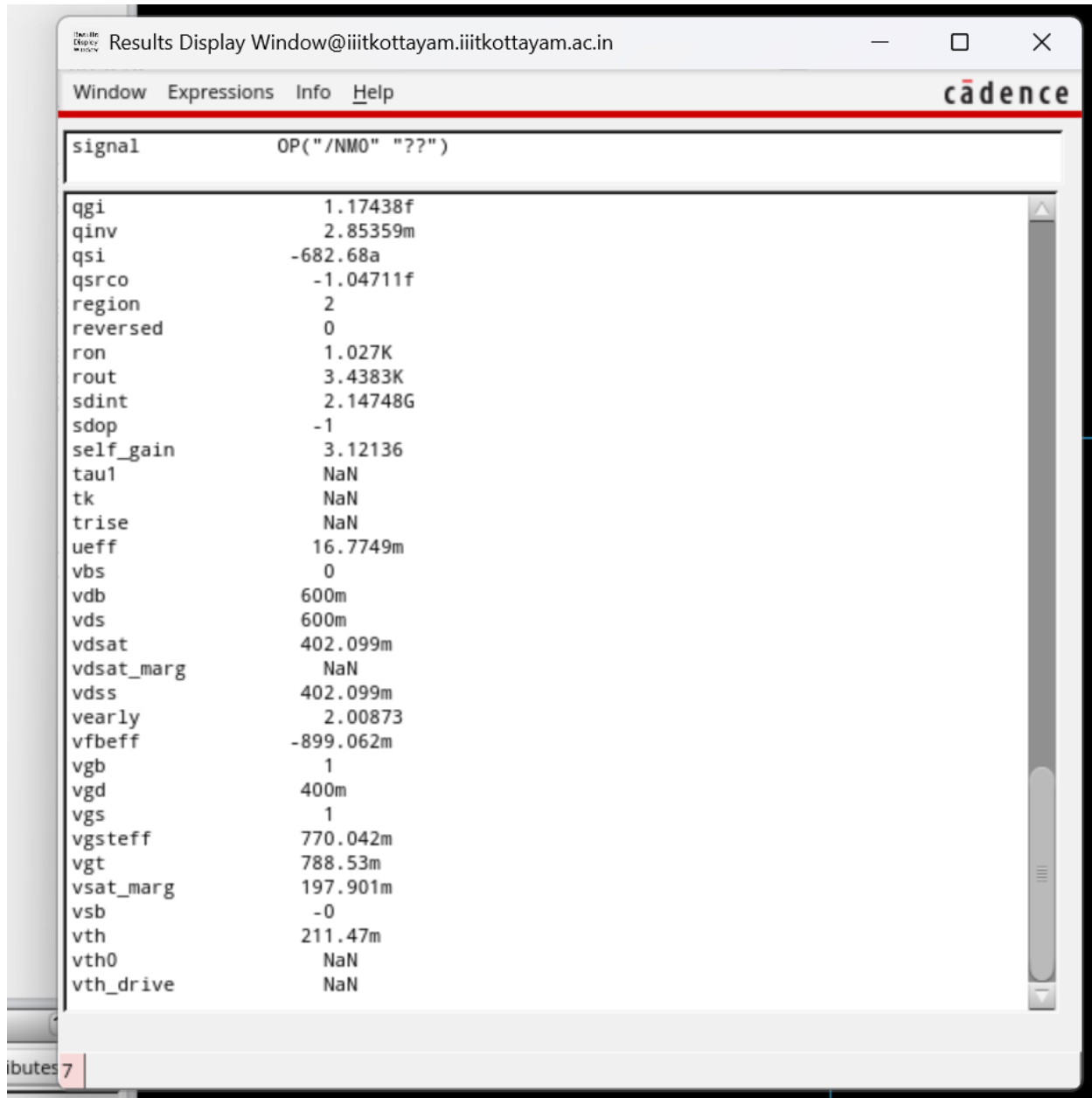


ADE L setup showing the DC analysis and design variables DS and GS.



DC operating-point analysis selection with VDS configured as the swept design variable.

DC Operating Point



The screenshot shows the 'Results Display Window' in Cadence, displaying a table of DC operating point parameters. The window title is 'Results Display Window@iitkottayam.iitkottayam.ac.in'. The table lists various parameters and their values, such as q_{gi} , q_{inv} , q_{si} , q_{srco} , $region$, $reversed$, ron , $rout$, $sdint$, $sdop$, $self_gain$, $tau1$, tk , $trise$, $ueff$, vbs , vdb , vds , $vdsat$, $vdsat_marg$, $vdss$, $vearly$, $vfbeff$, v_{gb} , v_{gd} , v_{gs} , v_{gsteff} , v_{gt} , v_{sat_marg} , v_{sb} , v_{th} , v_{th0} , and v_{th_drive} .

signal	OP("/NM0" "??")
q_{gi}	1.17438f
q_{inv}	2.85359m
q_{si}	-682.68a
q_{srco}	-1.04711f
region	2
reversed	0
r_{on}	1.027K
r_{out}	3.4383K
$sdint$	2.14748G
$sdop$	-1
self_gain	3.12136
τ_1	NaN
t_k	NaN
trise	NaN
$ueff$	16.7749m
v_{bs}	0
v_{db}	600m
v_{ds}	600m
v_{dsat}	402.099m
v_{dsat_marg}	NaN
v_{dss}	402.099m
v_{early}	2.00873
v_{fbeff}	-899.062m
v_{gb}	1
v_{gd}	400m
v_{gs}	1
v_{gsteff}	770.042m
v_{gt}	788.53m
v_{sat_marg}	197.901m
v_{sb}	-0
v_{th}	211.47m
v_{th0}	NaN
v_{th_drive}	NaN

The operating-point results report the device parameters obtained from the Spectre simulation. For the displayed operating point, $V_{DS} = 0.6$ V and $V_{GS} = 1$ V; the reported threshold voltage is approximately 211.47 mV.